Evidence from Amazon Marketplace

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Roadmap (Talk-First Design)

1. Motivation & Background Facts
2. Model Intuition
3. Estimation Idea
4. Counterfactual
5. Results: How is Consumer Welfare Affected?
6. Interpretation

Roadmap

Question & Motivation

- **Question:** Does Amazon's vertical integration structure harm competition on its marketplace platform?
- Amazon is able to act as both a retailer and a platform for third-party sellers (3PS)
- What is the effect of this dual role as both competitor and marketplace operator on consumer welfare and competition?

Background Facts

- Amazon's share of total e-commerce sales was 40.4% in 2024
- Amazon experiences higher demand for its own products than 3PS do for theirs
  - 3PS sell 3 products per week
  - Amazon brands sell 221 per week
  - Amazon is only seller of $\approx$ half of the product it offers
- 3PS have two modes of fulfillment:
  - Fulfilled by Amazon: Amazon fulfills delivery for a fee
  - Fulfilled by Merchant: Cost of storage and delivery
- **What to do?**

Contribution

- Estimate a counterfactual in which Amazon is no longer a competitor
- Extend counterfactual to have a 3PS take Amazon's place
- Estimates demand, indirect utility, find bounds on fixed costs, structural separation

Paper's Story / Key Findings

- Counterfactual:
  - **No Re-entry:** Consumer welfare falls 18.6%, price increases by 1.7 pp., 20.6% profit increase for 3PS
  - **Re-entry:** Consumer welfare still falls, but up to 4.4 pp. less than in the above case
- Amazon as a seller appears to contribute value, thus implying the need for thought on the part of regulators

Roadmap

Setting: Amazon Marketplace

- Logistics: FBA vs FBM
- Referral Fees: A percentage of final price paid by consumer, typically 8%-15%
- Recommendation System:
  - Page Ranking: How a product is listed in search based on characteristics
  - Buy-Box: A group of sellers are listed on a page, but one is given more visibility
  - Frequently Bought Together: Grouped together by being bought together often

Data

- Keepa: Amazon scraper that provides product and offer characteristics for all listed products
  - Product Characteristics: title, brand, manufacturer, product description, real time changes in sales rank, product rating, number of reviews, Buy-Box sellers
  - Offer Characteristics: seller's name, logistic method, real time changes in prices/shipping costs
- AmzScout: market intelligence data utilized by 3PS
  - estimate agg. quantity sold from sales rank, keyword searches to compute the market

Data Cont.

- Focuses on wireless bluetooth headphones
  - Positive Sales
  - Product is a distinct barcode
- Assumes observed sales are only realized by the Buy-Box sellers, given most sales go through that channel

Descriptives

Table: Summary Statistics

Products	3688
Sellers	2253
% Producers	5.73
Amazon Brands	2
% 3PS FBA	78.79
% Using FBA and FBM	8.30

Descriptives Cont.

Table: Weekly Averages

Products	2345.07
Sellers	849
Sellers/Product	1.20
Products from 3PS	3.05
Products from Amazon	220.77
Price Amazon	156.20 USD
Price 3PS	80.90
Quantity Sold Amazon Retailer	49.45
Quantity Sold Amazon Brands	38.87
HHI	665

Descriptives Cont.

Table: Largest Sellers

Seller	Type	Products/Week
Amazon.com	Retailer/Producer	220.77
Sampell	Retailer	95.70
Zihnic	Producer	34.80
Thousandshores Inc	Retailer	30.30
Woot	Retailer	25.20
HEYE	Retailer	24.87
Avantree Direct USA	Producer	24.47
JLab Store	Producer	23.47
Trusted Kids Products	Retailer	23.33
Anker Direct	Producer	23.00

Roadmap

Model Setup

- Discrete choice demand model
- Products indexed by $j = 1, \ldots, J$ sold by sellers $r = 1, \ldots, R$ over weeks $t = 1, \ldots, T$ and months $m = 1, \ldots, M$
- 3PS pay fee ϕ dependent upon upon charged price, and additional fixed τ if using FBA
- FBM seller r has a product $j \in \mathcal{J}_{rt}$ with price p_{rt}
- FBA seller r has product $j \in \tilde{\mathcal{J}}_{rt}$ with price p_{rt}
- Sellers make two choices each week: set of products to offer and prices
- Set $\mathbf{J}_{rt} = \{\mathcal{J}_{rt}, \tilde{\mathcal{J}}_{rt}\}$ defines all products offered by seller r in week t

Demand

- Indirect utility of consumer i for product j in week t is:

$$U_{ijt} = \beta \mathbf{X}_{jrt} - \alpha_i p_{jrt} + \gamma_j + \gamma_m + \xi_{jrt} + \epsilon_{ijrt} \tag{1}$$

- $_{jrt}$: observed seller and product characteristics
- γ_j: product fixed effects
- γ_m: month fixed effects
- ξ_{jrt}: unobserved product characteristics
- ϵ_{ijrt}: i.i.d. extreme value error term s.t. outside option $U_{i0t} = \epsilon_{i0t}$

Demand Cont.

- Market share prediction is:

$$s_{jrt} = \int \frac{\exp\{\delta_{jrt} + \sigma\mu_i\}}{1 + \sum_{r=1}^{R} \sum_{j \in \mathcal{J}_{rt}} \exp\{\delta_{jrt}\} + \sum_{j \in \mathcal{J}_{at}} \exp\{\delta_{jat} + \sigma\mu_i\}} \, dF(\mu) \qquad (2)$$

- $\delta_{jrt} = \beta_{jrt} - \alpha p_{jrt} + \gamma_j + \gamma_m + \xi_{jrt}$
- $\sigma\mu_i$: random coefficient on price

Variable Profit

- Marginal cost for 3PS differs by fulfillment method:
  - FBM: c_{jrt}
  - FBA: $\tilde{c}_{jrt}$
- Firm variable profits:

$$\pi_{rt}(\mathcal{J}_{rt}, X_t) = \mathcal{M}_t \Bigg(\sum_{j \in \mathcal{J}_{rt}} s_{jrt}(p_t, X_t)\big[(1-\phi)p_{jrt} - c_{jrt}\big] \tag{3}$$

$$+ \sum_{j \in \tilde{\mathcal{J}}_{rt}} s_{jrt}(p_t, X_t)\big[(1-\phi)p_{jrt} - \tilde{c}_{jrt} - \tau\big] \Bigg) \tag{4}$$

Amazon Profit

- Amazon profit:

$$\pi_{at}(\mathcal{J}_{at}, X_t) = \mathcal{M}_t \Bigg(\sum_{j \in \mathcal{J}_{at}} s_{jat}(p_t, X_t) \left[p_{jat} - c_{jat} \right] \tag{5}$$

$$+ \sum_{r \neq a} \sum_{j \in \mathcal{J}_{rt}} s_{jrt}(p_t, X_t) \, \phi \, p_{jrt} + \sum_{r \neq a} \tau \cdot \mathbf{1}\{j \in \tilde{\mathcal{J}}_{rt}\} \Bigg) \tag{6}$$

Costs

- Marginal Costs are are derived from supply functions:

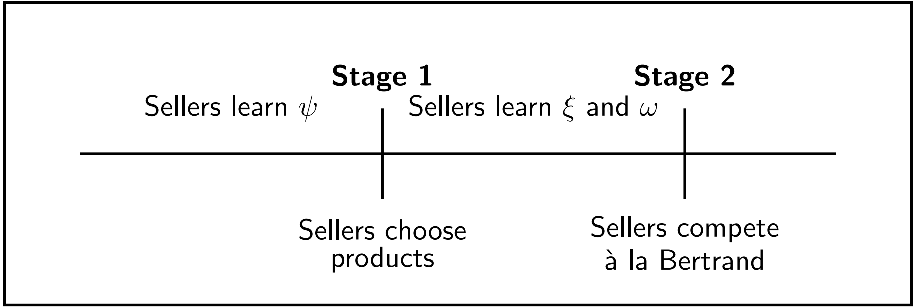
$$c_{jrt} = (\delta_1 + \delta_2 \mathbf{1}_{r=a})\mathbf{1}_{FBA} + \zeta_j + \zeta_t + \omega_{jrt} \tag{7}$$

- ζ_j: product fixed effects, ζ_t: week fixed effects, ω_{jrt}: error
- Fixed costs

$$f_{jrt} = \begin{cases} \theta_a + \psi_{jat}, & \text{if Amazon} \\ \theta + \psi_{jrt}, & \text{if 3PS using FBM} \\ \tilde{\theta} + \tilde{\psi}_{jrt}, & \text{if 3PS using FBA} \end{cases} \tag{8}$$

- ψ: product-seller specific unobserved fixed cost component
- $\tilde{\theta}$ takes into account opportunity cost of using FBA vs. other methods
- θ_a represents inventory costs and opportunity cost of selling for Amazon
- θ represents fixed costs for 3PS using FBM, including storage, refilling stock, etc.

Seller's Game



Timing of the Game

Roadmap

Estimation Overview

- Estimate demand via GMM using BLP framework, recover $(\beta, \alpha, \gamma, \sigma)$
  - Instruments: BLP and product characteristics of other products
- Recover marginal costs via first order conditions of profit maximization
- Back out bounds on fixed costs via entry/exit decisions (revealed preferences)

Roadmap

Demand

	Estimates		
	(1)	(2)	(3)
Intercept	-8.739 ***	-4.195 ***	-2.586 *
Price	-0.009 ***	-0.080 ***	-0.108 ***
Amazon	1.483 ***	1.409 ***	1.374 ***
Fulfilled by Amazon (FBA)	0.438 ***	0.378 ***	0.395 ***
Product Rating$_{\text{week-1}}$	0.167 ***	0.229 ***	0.299 ***
Log Reviews$_{\text{week-1}}$	0.079 ***	0.069 ***	0.083 ***
σ			0.018 **
			(0.007)
Product FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
IV	No	Yes	Yes
Model	Logit	Logit	RC Logit
Median η_{own}	-0.402	-3.653	-4.631
Mean η_{own}	-0.762	-6.924	-7.292

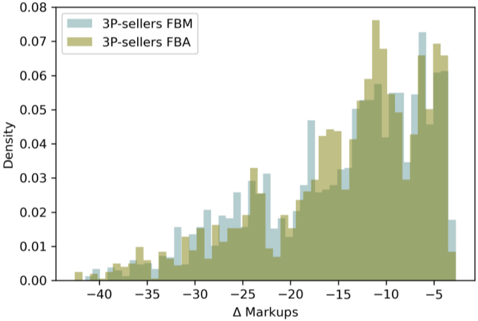
Cross-Price Elasticities

	Seller – Product – Brand							
	r1-j1-Sony	r2-j2-PowerLocus	r3-j3-QearFun	r4-j4-Perytong	r5-j5-Jabra	a-j6-Beats	a-j7-Belkin	a-j8-Kef
r1-j1-Sony	-10.837	0.001	0.001	0.001	0.003	0.003	0.001	0.003
r2-j2-PowerLocus	0	-3.178	0	0	0	0	0	0
r3-j3-QearFun	0	0	-2.497	0	0	0	0	0
r4-j4-Perytong	0	0.001	0.001	-2.099	0	0	0.001	0
r5-j5-Jabra	0.010	0.002	0.002	0.002	-14.772	0.025	0.002	0.026
a-j6-Beats	0.030	0.003	0.003	0.002	0.067	-20.348	0.004	**0.131**
a-j7-Belkin	0	0	0	0	0	0	-3.429	0
a-j8-Kef	0.001	0	0	0	0.003	0.006	0	-26.371

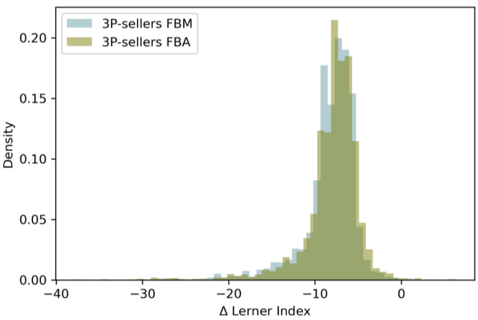
Marginal Costs

Variable	Estimates
FBA	-3.99 ***
Amazon	7.13 ***
Product FE	Yes
Week FE	Yes
Observations	84149

Markups and Lerner Index



(a) Markups



(b) Lerner Index

Fixed Costs Bounds

$\hat{\theta}$	Estimated Bounds (USD)	95% C.I.
θ	[6, 45]	[5, 50]
$\tilde{\theta}$	[15, 76]	[12, 80]
θ_a	[18, 244]	[8, 272]

Roadmap

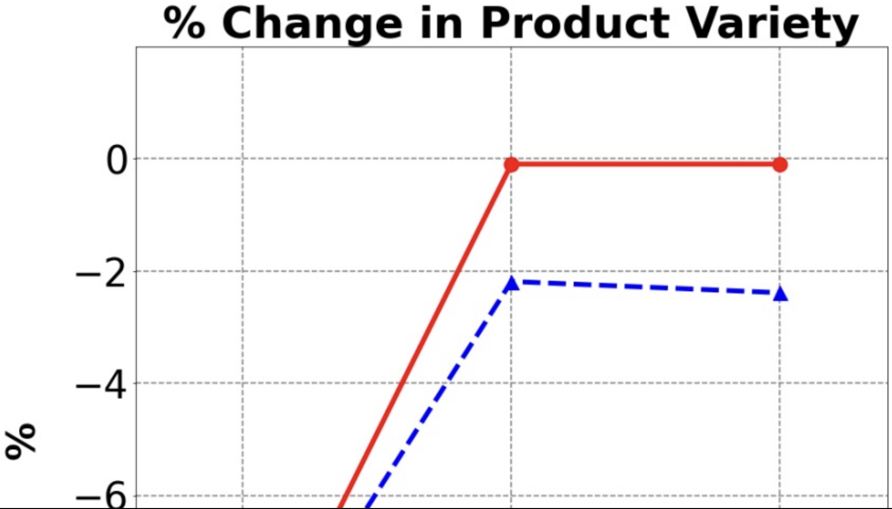
Counterfactual Setup

- Remove Amazon as a seller
- Two scenarios:
  - No Re-entry: 3PS cannot enter to fill Amazon's spot
  - Re-entry: A new 3PS enters to fill Amazon's spot
- Solve for new equilibrium prices and quantities
- Compute consumer welfare changes via compensating variation

Counterfactual Results

	Basis (US$)	No Entry ($\%\Delta$)	No Entry ($\%\Delta$)	FBM Entrant [min $\%\Delta$, max $\%\Delta$]	FBA Entrant [min $\%\Delta$, max $\%\Delta$]
Consumer Surplus	28'530'493	-18.57	-3.48	[0.04, 0.16]	[1.76, 1.95]
Amazon					
Amazon Profit	20'487'548	-25.69	-38.75	[-36.77, -36.70]	[-34.27, -34.11]
Amazon Fees Revenues	12'023'255	26.62	4.37	[7.75, 7.86]	[12.00, 12.28]
Incumbent non-Amazon Sellers					
Variable Profit	10'270'051	20.60	46.92	[41.65, 41.87]	[39.29, 39.64]
Prices	81	0.37	-4.91	[-5.0, -5.0]	[-5.1, -5.08]
Amazon-sold products pre-ban					
Incumbent non-Amazon Sellers Prices	0	0.51	-4.21	[-4.27, -4.26]	[-4.29, -4.28]
Entrant Prices	0	0	0	[-6.04, -5.06]	[-7.11, -5.94]

Counterfactual Results



Roadmap

Takeaways

- Amazon's dual role as both a retailer and a platform operator has significant effects on competition and consumer welfare
- Counterfactual analysis shows that removing Amazon as a seller leads to a decrease in consumer welfare, even with potential re-entry by other sellers
- The presence of Amazon as a seller appears to contribute value to the marketplace, suggesting that regulators should carefully consider the implications of vertical integration in e-commerce platforms

Thanks!

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